THE ADIC COMPLETION OF A FLAT MODULE IS FLAT MITTAG-LEFFLER WHEN THE REDUCTION MODULO THE IDEAL IS PROJECTIVE OVER A COMPLETE NOETHERIAN BASE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Splitting of the completed sequence, exhibiting the completion as a direct summand. Algebra, label algebra-lemma-split-completed-sequence (invoked at proof line 2045). For a surjection $F \to M$ with F free, it makes $M^\wedge$ a direct summand of $F^\wedge$, so the claim reduces to the free case.
- Flatness and the Mittag-Leffler condition for the completion of a free module. Same chapter flat.tex, label lemma-completed-direct-sum-ML (invoked at proof line 2048). It settles the case $M = F$ free, which is what the reduction requires.

Source reference. Original lemma 05D3: sources/stacks-chow/tags.txt, source lines 6941–6941. Navigation pointer for \ref{algebra-lemma-split-completed-sequence} used inside the counted span (line 2045); the argument that consumes it is retained in the span.

Source reference. Original lemma 05DJ: sources/stacks-flat/pinned-original-source.tex, source lines 2003–2003. Navigation pointer for \ref{lemma-completed-direct-sum-ML} used inside the counted span (line 2049); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a ring. Let $I \subset R$ be an ideal. Let M be an R-module. Assume*

(1) *R is Noetherian and I-adically complete,*

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(2) *M is flat over R, and*
(3) *M/IM is a projective R/I-module.*

Then the I-adic completion $M^\wedge$ is a flat Mittag-Leffler R-module.

Proof. Choose a surjection $F \to M$ where F is a free R-module. By Algebra, Lemma 05D3 the module $M^\wedge$ is a direct summand of the module $F^\wedge$. Hence it suffices to prove the lemma for F. In this case the lemma follows from Lemma 05DJ. □